BALANCING REPRESENTATIONS BY IDENTIFYING AND GENERATING UNDERREPRESENTED DATA

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ABSTRACT

Self-supervised learning inherits gaps in its source data. We study whether these gaps can be repaired while representation learning is in progress. We introduce BRIDGE, a label-free feedback loop in which the current SSL representation scores local support, a budgeted acquisition policy selects sparse but recoverable regions, and a frozen generator adds local variants before continued pretraining. The resulting intervention changes the representation that determines the next intervention. Across five long-tailed and web-source regimes, BRIDGE improves average linear-probe and kNN transfer over source-matched SSL baselines. Paired full-schedule experiments further show gains with both ResNet-50 and ViT-S, while low-label fine-tuning and semantic segmentation extend the evidence beyond frozen-feature evaluation. Selection and repair controls compare mode-window acquisition with uniform sampling, extreme-tail selection, real-data restoration, and matched generator settings. We formalize the acquisition problem through recoverability-weighted marginal support utility. The analysis favors an interior repair region when the value of added support grows with sparsity but generator reliability falls in the extreme tail, and it favors adaptive reallocation when regional gains diminish after repair. Overall, the results show that allocating a limited data budget from an evolving representation can improve learning from skewed unlabeled data.

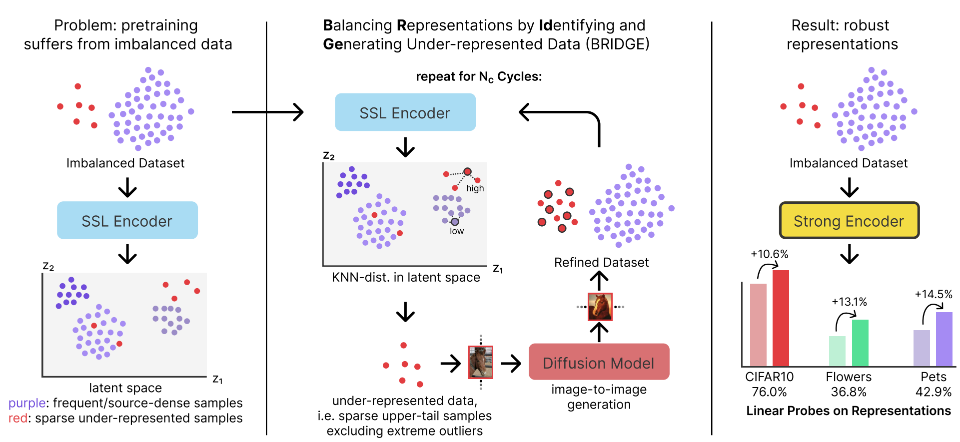

Figure 1: Conceptual overview of BRIDGE. We alternate between SSL pretraining, sparsity scoring in representation space, and targeted image-to-image diffusion augmentation. BRIDGE augments a mode-centered band inside the upper OOD tail rather than the extreme tail itself.

1 INTRODUCTION

Self-supervised learning is now a standard way to learn transferable visual representations from unlabeled images. Strong methods span contrastive learning, masked prediction, and self-distillation (Chen et al., 2020; He et al., 2020; 2022; Caron et al., 2021; Oquab et al., 2024; Siméoni et al., 2025). Yet these methods still inherit coverage biases from the source corpus. Prior work shows that imbalance can hurt self-supervised learning and motivates specialized long-tailed training strategies (Tian et al., 2021; Assran et al., 2023; Kukleva et al., 2023; Jiang et al., 2021). Learned features can concentrate around frequent modes while providing weaker coverage of underrepresented regions. The experiments in Section 6 test whether adding data to such regions improves transfer.

Most prior attempts to address long-tailed SSL are model-centric. Temperature Schedules reshapes the optimization trajectory of contrastive learning over time (Kukleva et al., 2023). SDCLR introduces a self-competitor to emphasize harder and rarer samples (Jiang et al., 2021). These methods are strong baselines, but they still treat the source distribution as largely fixed.

We study the complementary direction and ask whether we can repair the source data while SSL training proceeds. We propose BRIDGE, a cyclic data-repair loop shown in Figure 1. After each pretraining stage, we embed the current training set, score each image by mean kNN distance, select a sparse but still semantically recoverable region of representation space, and generate new training images for that region with a frozen diffusion model. The next cycle then trains on the repaired dataset. In effect, the encoder reveals where the source manifold is thin, and the generator densifies those regions before the next round of learning.

Two aspects of BRIDGE are central. First, BRIDGE restricts attention to the upper sparse band of the OOD distribution, finds the mode within that band, and then selects a diverse set of anchors around that mode. This avoids the super-OOD tail, which often contains noisy or weakly useful images, while retaining the underrepresented region that is most amenable to repair. In the ablations, we compare this selector against top-tail and uniform alternatives. Second, we evaluate BRIDGE across five distinct source regimes. In addition to ImageNet-100-LT, CIFAR-10-LT, and CIFAR-100-LT, we study web-source unlabeled data drawn from PASS and DiffusionDB. Across these five regimes, BRIDGE is best on the seven-task average and remains strong under both linear-probe and kNN transfer.

We make three contributions. First, we formulate data repair as recoverability-weighted support allocation and derive when a finite budget should favor sparse interior regions and adaptive reallocation. BRIDGE gives this formulation a label-free cyclic implementation. Second, we introduce an upper-tail mode-window rule with diversity-aware subsampling, which targets semantically recoverable sparse regions without drifting into the extreme OOD tail or wasting budget on near-duplicates. Third, we evaluate five source regimes and seven transfer tasks, with paired ViT-S experiments and downstream tests that include full fine-tuning and semantic segmentation.

2 Background and Related Work

Skewed SSL and representation-driven data selection. Long-tailed SSL methods modify training while leaving the source set fixed. SDCLR emphasizes persistently difficult samples, Temperature Schedules changes contrastive optimization, and COLT introduces external OOD data (Jiang et al., 2021; Kukleva et al., 2023; Bai et al., 2023). Other studies examine hidden imbalance, frequency-aware learning, and uncurated data (Tian et al., 2021; Assran et al., 2023; Lin et al., 2023). Data curation uses representations to retain informative subsets or remove redundancy (Whang et al., 2023; Sorscher et al., 2022; Abbas et al., 2024). Earlier augmentation methods instead add support. SMOTE interpolates labeled minority examples, while DOPING oversamples low-density latent regions for unsupervised anomaly detection (Chawla et al., 2002; Lim et al., 2018). BRIDGE uses the current SSL representation to decide where new source images should be added, then recomputes that decision after learning from them.

Generative augmentation. Diffusion models make image-conditioned repair possible through controlled denoising (Ho et al., 2020; Rombach et al., 2022; Meng et al., 2022). DA-Fusion applies this mechanism to few-shot supervised recognition, and TADA targets examples learned slowly under a supervised objective (Trabucco et al., 2024; Nguyen et al., 2026). Larger studies show that synthetic-data utility depends on prompt design, filtering, diversity, generator configuration, and domain gap (Azizi et al., 2023; He et al., 2023; Fan et al., 2024). BRIDGE uses neither class identity nor supervised learning speed. It allocates a fixed generation budget from local support in an unlabeled representation.

Synthetic representation learning and data engines. Synthetic imagery can serve as the full training source or define contrastive views (Ren and Lee, 2018; Baradad et al., 2021; 2022; Jahanian et al., 2022; Sariyildiz et al., 2023; Tian et al., 2023). DiffAug is the closest iterative SSL method. It jointly learns a conditional generator for positive views, while BRIDGE holds the generator fixed and studies where it should intervene in an observed source corpus (Zang et al., 2024). AIDE and industrial data engines also close a model–data loop, but identify supervised task failures and use language models, annotation, and domain-specific verification (Liang et al., 2024; Worker and Pathak, 2022). We evaluate language-mediated acquisition and text-to-image generation as direct alternatives.

External representation priors. DreamTeacher distills generative features into discriminative backbones, DIFT extracts semantic diffusion features, and REPA transfers visual features in the opposite direction into a diffusion model (Li et al., 2023; Tang et al., 2023; Yu et al., 2025). A frozen generator therefore contributes knowledge from external pretraining even when its parameters never change. Our controls compare direct feature transfer, uniform generation, and targeted generation from the same model. The remaining question is how an unlabeled learner should allocate a limited intervention budget as its representation evolves. Section 3 formulates this problem.

3 Theoretical Framework

We formulate targeted data repair as a budgeted allocation problem over coherent regions of representation space. A region is valuable when additional support can reduce its contribution to downstream risk, while a repair is useful only when the generator preserves the semantic content of its anchor. The acquisition policy must therefore balance support deficit, repairability, and redundancy.

Let $r_k(z; Z)$ be the distance from z to its kth neighbor in representation set Z. Let A be a set of valid generated variants associated with z.

Proposition 1 (Local support monotonicity). *For any A, $r_k(z; Z \cup A) \leq r_k(z; Z)$. The inequality is strict whenever an added point enters the first k neighbors and displaces a point at strictly larger distance.*

The statement follows because enlarging the candidate set cannot increase its kth order statistic. To connect support to learning, partition representation space into coherent regions j with current effective support n_j. We model a regional learning curve by

$$\mathcal{B}(\mathbf{m}) = \sum_j p_j a_j (n_j + m_j + \kappa)^{-\gamma}, \tag{1}$$

where p_j is the transfer-relevant mass of region j, a_j is its difficulty, m_j is the number of valid repairs allocated to it, $\kappa > 0$, and $\gamma > 0$. The marginal reduction from one valid repair is

$$\Delta_j(m_j) = p_j a_j \left[(n_j + m_j + \kappa)^{-\gamma} - (n_j + m_j + 1 + \kappa)^{-\gamma}\right]. \tag{2}$$

Proposition 2 (Sparse-region marginal value). *$\Delta_j(m_j)$ is positive and strictly decreases with $n_j + m_j$. When two regions have equal transfer mass and difficulty, the region with less effective support has the larger marginal value.*

This follows from strict monotonicity and convexity of $x \mapsto x^{-\gamma}$. It provides a direct reason to allocate a limited budget away from regions that are already well supported.

Generation can fail, especially for corrupt or semantically incoherent anchors. Write d for a sparsity score, $b(d)$ for marginal value conditional on valid repair, $\rho(d)$ for the probability of a faithful and useful variant, and $\lambda \geq 0$ for the cost of an invalid variant. Expected repair utility is

$$u(d) = \rho(d)b(d) - \big(1 - \rho(d)\big)\lambda. \tag{3}$$

Proposition 3 (Interior recoverable-support optimum). *Suppose b and ρ are positive and differentiable on $[d_{\min}, d_{\max}]$, with $b'(d) > 0$ and $\rho'(d) < 0$. Define*

$$g(d) = \frac{b'(d)}{b(d) + \lambda} + \frac{\rho'(d)}{\rho(d)}. \tag{4}$$

If g is strictly decreasing, $g(d_{\min}) > 0$, and $g(d_{\max}) < 0$, then u has a unique maximizer $d^\star$ in the interior, characterized by $g(d^\star) = 0$.

The condition compares the relative growth of support value with the relative decay of repairability. Dense regions are preferred while value grows faster, and utility begins to fall once reliability decays faster. The maximizing score depends on the source distribution and generator through b and ρ. BRIDGE operationalizes this interior preference with a trimmed score band, while the experiments test lower and upper cutoffs, k, distance normalization, candidate-pool size, and diversity sampling.

Finally, consider a separable budget objective $U(\mathbf{m}) = \sum_j U_j(n_j + m_j)$, where each U_j is increasing and discretely concave and $\sum_j m_j = B$.

Proposition 4 (Adaptive allocation under diminishing returns)**.** *For integer allocations, repeatedly assigning one repair to the region with the largest current marginal gain maximizes $U(\mathbf{m})$. A fixed allocation can become suboptimal after a region's marginal gain falls below that of another region.*

The result follows by selecting the B largest elements from the nonincreasing marginal-gain sequences of all regions. Recomputing the acquisition signal lets the policy follow these changing values, whereas a frozen policy continues to spend according to its initial ordering. Within a candidate region, farthest-first traversal gives a factor-two approximation to metric k-center coverage (Gonzalez, 1985), providing a coverage interpretation for diversity-aware selection.

The analysis yields four requirements for a practical acquisition policy. It should estimate local support, avoid sparse regions whose samples cannot be repaired faithfully, cover the chosen region without redundant anchors, and recompute its allocation after learning from each intervention. Appendix C proves Propositions 1–4. Equation 1 and the monotonic behavior assumed for b and ρ are modeling assumptions rather than consequences of those proofs. Their observable predictions are tested by the selection, repair, cycle, and policy controls in Section 6 and the appendix. The next section turns the framework into a label-free algorithm.

4 BRIDGE

4.1 Cyclic acquisition and continued pretraining

BRIDGE instantiates each requirement using quantities available without labels. Mean kNN distance estimates local support. An upper-tail mode window seeks a sparse region while excluding the most extreme scores, where faithful repair is less reliable. Farthest-point sampling limits redundancy within this region, and image-to-image diffusion supplies local variants. Continued SSL changes the representation, after which BRIDGE estimates support and allocates the next budget again.

Let $D^{(0)} = \{x_i\}_{i=1}^{N_0}$ denote an unlabeled source dataset and let f_θ be an SSL encoder trained with objective $\mathcal{L}_{\text{ssl}}$. BRIDGE alternates between pretraining and dataset repair over C cycles. At cycle c, the current encoder is $f_{\theta^{(c)}}$, the current source set is $D^{(c)} = \{x_i\}_{i=1}^{N^{(c)}}$, and the corresponding embedding set is $Z^{(c)} = \{z_i^{(c)}\}_{i=1}^{N^{(c)}}$.

The loop applies three operators after each SSL training stage, which score the data, select anchors, and repair the source set. The score operator embeds all current source images and assigns an unlabeled sparsity score. The select operator converts that score distribution into a finite anchor set under a generation budget. The repair operator uses those anchors as image-to-image conditioning inputs and appends the generated variants to the next-cycle source set. The SSL encoder is the only learned component in this loop. The OOD scoring rule is nonparametric, the selection rule has no trainable parameters, and the image generator is frozen. This interface keeps BRIDGE independent of class labels and makes it compatible with any SSL objective that can provide image embeddings.

After training on $D^{(c)}$, we embed every image and compute a nonparametric OOD score from mean kNN distance,

$$\begin{aligned} z_i^{(c)} &= f_{\theta^{(c)}}(x_i), \\ d_i^{(c)} &= \frac{1}{k} \sum_{z \in \mathcal{N}_k\left(z_i^{(c)};\, Z^{(c)} \setminus \{z_i^{(c)}\}\right)} \left\| z_i^{(c)} - z \right\|_2^2, \end{aligned} \tag{5}$$

where $\mathcal{N}_k(z; Z)$ denotes the set of the k nearest neighbors of z in Z. Deep nearest-neighbor distance is also effective for OOD detection (Sun et al., 2022). Here, large $d_i^{(c)}$ indicates that an image lies in a sparse part of the learned representation space.

4.2 SPARSE AND RECOVERABLE ANCHOR SELECTION

A natural alternative is to select the largest OOD scores directly. In practice, that rule is unstable. The farthest samples are often artifacts, degenerate crops, or semantically incoherent outliers. Uniform sampling instead spreads budget across the whole dataset, including regions that are already well covered. We target a sparse region that is still internally coherent.

Our selector works in three steps. First, it restricts attention to the upper sparse band of the empirical OOD distribution, namely the scores between the 75th and 99th percentiles, and builds a histogram over that band. Let $b^\star$ denote the modal bin and let $m^{(c)}$ denote the center of that bin. Second, given an augmentation budget B and a candidate-pool multiplier α, we form a mode-centered candidate set of size $M = \alpha B$:

$$C^{(c)} = \operatorname*{arg\,min}_{C \subseteq U^{(c)},\, |C| = M} \sum_{i \in C} \left| d_i^{(c)} - m^{(c)} \right|, \tag{6}$$

where $U^{(c)} = \{i : q_{0.75}^{(c)} \le d_i^{(c)} \le q_{0.99}^{(c)}\}$ and $[N^{(c)}] = \{1, \dots, N^{(c)}\}$. Third, we choose the final anchor set $S^{(c)}$ by greedy farthest-point sampling in embedding space from that candidate pool:

$$S^{(c)} = \mathrm{FPS}\Big(C^{(c)}, B; Z^{(c)}\Big). \tag{7}$$

This keeps selection inside a stable sparse regime while discouraging near-duplicate anchors. In our implementation we use $\alpha = 4$, so the final anchors are chosen from a candidate pool that is both mode-centered and genuinely sparse. Appendix Figure 2 reports how the acquisition distribution changes across repair cycles.

4.3 LOCAL GENERATIVE REPAIR

Given the selected set $S^{(c)}$, we generate G synthetic variants per selected image with a frozen image-to-image diffusion model g_ϕ. Each selected source image is converted back to image space and used as the conditioning image. We use an empty text prompt so that the repair signal comes from the selected anchor rather than from class names or hand-written prompts. In the main method we use Stable Diffusion 3 Medium in image-to-image mode with 20 denoising steps, guidance scale 5.0, strength 0.6, and output resolution equal to the training crop size. The generation ablation replaces this operator with FLUX, using six steps and guidance scale 2.5, or with a no-generation re-population baseline that re-adds the originally removed source images without diffusion. We first form the generated set

$$\begin{aligned} \widetilde{D}^{(c)} &= \Big\{ g_\phi(x_i, \epsilon_j) \,\Big|\, i \in S^{(c)},\, j \in [G] \Big\}, \\ D^{(c+1)} &= D^{(c)} \cup \widetilde{D}^{(c)}, \end{aligned} \tag{8}$$

where $[G] = \{1, \dots, G\}$. The random noise input ϵ_j indexes independent translations of the same anchor, so multiple generated images can preserve coarse visual structure while changing texture, pose, color, or local composition. We do not apply an additional generated-image filter in the main pipeline.

Together, these components implement the allocation principle from Section 3. The score estimates support deficit, the mode window provides a practical recoverability constraint, FPS approximates coverage under the finite anchor budget, and cyclic re-estimation updates the allocation after every intervention. The experiments test these choices against uniform and extreme-tail selection, alternative repair operators, and matched generator settings.

5 EXPERIMENTAL SETUP

Source datasets. We pretrain on five unlabeled source regimes. Three are long-tailed image classification datasets, namely ImageNet-100-LT (Liu et al., 2019), CIFAR-10-LT, and CIFAR-100-LT (Krizhevsky, 2009). Two are web-source unlabeled corpora, namely PASS (Asano et al., 2021) and DiffusionDB (Wang et al., 2023). For PASS and DiffusionDB we use 10k example subsets.

Downstream evaluation. We evaluate transfer on seven benchmarks, comprising CIFAR-10-LT, CIFAR-100-LT, Stanford Cars (Krause et al., 2013), FGVC Aircraft (Maji et al., 2013), Oxford Flowers (Nilsback and Zisserman, 2008), Oxford-IIIT Pets (Parkhi et al., 2012), and ImageNet-100-LT. In the main text, we report full linear-probe comparisons, together with kNN corroboration for the source-regime comparisons and the combined ablation table. The appendix provides the complete expanded linear-probe and kNN tables for every setting.

Training protocol. Unless stated otherwise, reported numbers are mean $\pm$ standard deviation over three seeds. We use ResNet-50 as the default backbone because it is the common architecture across the SSL and long-tailed SSL baselines we compare against, enabling controlled method-level comparisons without confounding improvements from newer backbone families. We also include an architecture ablation over ResNet-18, ResNet-50, ViT-S, and ViT-B. Unless noted otherwise, BRIDGE uses $k = 100$ for OOD scoring, selects 500 images per cycle, generates 5 images per selected image, and trains for 5 cycles with 100 epochs per cycle. The cycle ablation changes the number of cycles while adjusting the per-cycle budget accordingly. The successful DINO ablation uses two local crops. Because the downstream tasks are heterogeneous, we emphasize per-dataset tables and use the average only as a compact summary statistic. Appendix D gives additional implementation details, runtime summaries, and asset notes.

Compared methods. We compare SimCLR, TS, SDCLR, and BRIDGE. Hybrid results that combine BRIDGE with TS or SDCLR are reported in the appendix. We also pair BRIDGE with SimCLR and MoCo v3 on ViT-S to separate backbone and objective effects.

Ablation protocol. All ablations are run on ImageNet-100-LT source pretraining with the same downstream suite and three-seed evaluation. The pretraining ablation swaps SimCLR, MoCo, and DINO. The generation ablation compares Stable Diffusion 3, FLUX, and no-generation re-population of the originally removed data. This sanity check isolates the effect of semantically guided generation from the effect of merely increasing the amount of data. The selection ablation compares three upper-tail mode-window ranges, a top-tail rule that takes the most OOD examples directly, and uniform sampling. The cycle ablation varies the number of repair cycles. The architecture ablation compares ResNet-18, ResNet-50, ViT-S, and ViT-B.

6 RESULTS

6.1 COMPARISON ACROSS SOURCE DATASETS

Table 1 reports the three label-derived long-tailed source regimes with all seven downstream tasks shown directly. BRIDGE is the strongest method on the overall average in all three regimes. The clearest case is CIFAR-100-LT source pretraining, where BRIDGE raises the seven-task average from 23.8 for source-matched SimCLR to 34.4 and is best on every downstream dataset. On ImageNet-100-LT, BRIDGE improves the average from 39.8 to 42.5 and remains strongest overall even though TS stays competitive on the in-domain ImageNet-100-LT column. On CIFAR-10-LT, BRIDGE improves the average from 29.1 to 34.2 and is again best overall. Across the three regimes, objective-level changes can help selectively, but repairing the source set produces the strongest broad transfer.

The appendix shows that this ranking is not specific to linear probing. Under kNN transfer, BRIDGE remains strongest on average in all three label-derived regimes, improving over SimCLR from 31.8 to 33.6 on ImageNet-100-LT, from 24.0 to 28.7 on CIFAR-10-LT, and from 20.9 to 27.0 on CIFAR-100-LT.

Table 2 (linear-probe) evaluates the two web-source regimes, which are qualitatively different from the label-derived long-tailed datasets. PASS is a natural web image corpus, while DiffusionDB is fully synthetic. BRIDGE is again the strongest average method in both settings. On PASS-10k, it improves the seven-task average over source-matched SimCLR from 36.8 to 39.9. On DiffusionDB-10k, it improves the average from 37.7 to 40.2. The DiffusionDB result is especially informative. Because the source data are already synthetic, the gains cannot be attributed only to harvesting more natural web diversity. They instead indicate that the repair loop remains useful even when the source corpus itself carries strong generative and stylistic bias.

Table 1: Linear-probe transfer for the three label-derived long-tailed source regimes. The average is computed over all seven downstream linear-probe tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 ± 0.5	46.8 ± 0.4	13.7 ± 0.3	19.6 ± 0.2	35.9 ± 7.2	38.6 ± 1.5	50.4 ± 0.5	39.8 ± 1.5
	TS	72.5 ± 0.4	44.0 ± 0.5	12.5 ± 1.2	15.0 ± 0.1	35.7 ± 1.8	40.6 ± 2.2	**56.7 ± 1.2**	39.6 ± 1.1
	SDCLR	41.1 ± 1.7	18.0 ± 0.6	1.4 ± 0.3	2.3 ± 0.5	10.8 ± 1.7	10.4 ± 0.3	21.9 ± 0.6	15.1 ± 0.8
	BRIDGE (ours)	**75.8 ± 0.2**	**49.3 ± 0.2**	**16.4 ± 1.2**	**22.4 ± 0.4**	**36.8 ± 1.9**	**42.9 ± 1.0**	53.8 ± 0.1	**42.5 ± 0.7**
CIFAR-10-LT	SimCLR	67.8 ± 0.2	38.0 ± 1.0	6.5 ± 0.4	16.0 ± 0.9	**19.2 ± 1.6**	24.0 ± 1.3	32.0 ± 0.7	29.1 ± 0.9
	TS	68.5 ± 1.3	36.0 ± 1.7	4.3 ± 0.2	10.3 ± 0.1	13.1 ± 2.0	19.7 ± 2.3	31.5 ± 1.2	26.2 ± 1.3
	SDCLR	39.6 ± 1.1	15.6 ± 0.2	1.1 ± 0.1	2.4 ± 0.9	12.0 ± 1.6	9.1 ± 1.8	16.5 ± 1.6	13.8 ± 1.1
	BRIDGE (ours)	78.1 ± 0.3	**49.8 ± 0.4**	**9.2 ± 0.8**	**17.7 ± 1.0**	18.2 ± 1.1	**27.4 ± 2.7**	39.0 ± 0.6	**34.2 ± 1.0**
CIFAR-100-LT	SimCLR	59.0 ± 1.5	31.1 ± 1.6	4.1 ± 0.3	11.2 ± 1.0	12.6 ± 3.4	20.1 ± 1.0	28.7 ± 1.3	23.8 ± 1.4
	TS	56.3 ± 0.2	28.0 ± 0.1	2.8 ± 0.6	8.5 ± 0.2	15.5 ± 2.5	16.2 ± 3.1	26.2 ± 1.8	21.9 ± 1.2
	SDCLR	37.1 ± 0.8	15.7 ± 0.2	1.9 ± 0.1	2.4 ± 0.7	9.4 ± 1.0	9.9 ± 1.7	16.2 ± 0.3	13.2 ± 0.7
	BRIDGE (ours)	**74.2 ± 0.5**	**48.1 ± 0.4**	**8.3 ± 1.0**	**18.2 ± 0.7**	**24.4 ± 2.3**	27.6 ± 0.7	39.8 ± 1.1	**34.4 ± 1.0**

Table 2: Linear-probe transfer for the two web-source regimes. PASS is a natural web-image corpus, while DiffusionDB is fully synthetic. The average is computed over all seven downstream linear-probe tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
PASS-10k	SimCLR	72.3 ± 0.2	45.5 ± 0.5	11.0 ± 0.2	17.5 ± 1.4	33.3 ± 1.3	33.3 ± 0.9	44.7 ± 0.9	36.8 ± 0.8
	TS	71.5 ± 0.2	43.5 ± 0.4	10.2 ± 1.2	17.2 ± 0.2	33.1 ± 2.1	36.4 ± 1.8	49.4 ± 0.4	37.3 ± 0.9
	SDCLR	42.2 ± 1.1	17.7 ± 0.8	1.8 ± 0.3	2.2 ± 0.8	10.6 ± 1.3	8.2 ± 1.6	20.8 ± 0.4	14.8 ± 0.9
	BRIDGE (ours)	**73.9 ± 0.3**	**47.1 ± 0.4**	**13.6 ± 0.7**	**20.5 ± 0.7**	**37.7 ± 4.0**	**37.1 ± 0.7**	49.1 ± 0.5	**39.9 ± 1.0**
DiffusionDB-10k	SimCLR	74.4 ± 0.5	48.6 ± 0.2	10.9 ± 0.7	17.6 ± 0.8	31.2 ± 2.4	36.1 ± 1.3	45.3 ± 0.5	37.7 ± 0.9
	TS	72.3 ± 0.3	44.4 ± 0.3	10.7 ± 0.6	15.4 ± 0.6	33.1 ± 1.1	**39.9 ± 2.7**	48.8 ± 0.2	37.8 ± 0.8
	SDCLR	42.4 ± 0.2	18.4 ± 0.2	1.1 ± 0.2	2.4 ± 0.1	13.0 ± 3.7	8.2 ± 1.4	21.0 ± 0.6	15.2 ± 0.9
	BRIDGE (ours)	**75.9 ± 0.4**	**49.9 ± 0.5**	**12.8 ± 1.0**	**20.0 ± 0.4**	**35.7 ± 1.1**	37.7 ± 0.8	49.1 ± 0.5	**40.1 ± 0.7**

The same conclusion holds under the appendix's nonparametric evaluation. On PASS-10k, BRIDGE improves the seven-task kNN average from 29.3 to 30.6. On DiffusionDB-10k, it improves the average from 28.2 to 29.9. Together with Table 2, this shows that the effect is not tied to a learned probe and not restricted to natural image corpora.

6.2 Evaluation beyond the default protocol

Table 3 tests whether the result is specific to SimCLR with ResNet-50 or to frozen-feature evaluation. In paired full-schedule experiments with ViT-S, BRIDGE improves the four-task average for both SimCLR and MoCo v3. Every paired run improves, although three pairs do not support a claim that the average MoCo v3 effect is statistically significant. DINO results are mixed and are reported in the appendix. We therefore claim compatibility with the completed SimCLR and MoCo v3 settings rather than uniform improvement across SSL objectives.

The same source representations improve after end-to-end adaptation. With 1 and 10 percent of the ImageNet-100-LT labels, full fine-tuning gains 1.98 and 3.16 points. A separate PASCAL VOC experiment gains 1.76 mIoU. The VOC result has one paired run, so it establishes task coverage rather than a variance estimate.

Table 3: Evaluation beyond the default SimCLR/ResNet-50 linear-probe setting. ViT-S results average Cars, Aircraft, Flowers, and ImageNet-100-LT over three paired runs. Fine-tuning uses the full encoder for 100 epochs over three paired runs. VOC segmentation is a single paired run and is reported without an uncertainty estimate.

Evaluation	Setting	SSL baseline	With BRIDGE	Difference
Frozen transfer	SimCLR, ViT-S	22.57 ± 0.94	24.58 ± 0.85	$+2.00 \pm 0.93$
Frozen transfer	MoCo v3, ViT-S	24.78 ± 0.77	26.16 ± 0.21	$+1.38 \pm 0.89$
Full fine-tuning	IN100-LT, 1% labels	14.31 ± 0.33	16.29 ± 1.11	$+1.97 \pm 0.78$
Full fine-tuning	IN100-LT, 10% labels	21.56 ± 0.75	24.72 ± 1.75	$+3.17 \pm 1.66$
Segmentation	PASCAL VOC, mIoU	10.56	12.32	$+1.76$

6.3 Ablations

We now examine each design choice separately on ImageNet-100-LT source pretraining. Each ablation changes one component of BRIDGE while holding the rest of the pipeline fixed. We summarize the principal findings below and provide complete per-task linear-probe and kNN tables in the appendix. The goal is to identify which parts of the repair loop matter most, and whether the strongest configuration is driven by one isolated choice or by a coherent combination of choices.

SSL objective and backbone. The ResNet-50 objective swap in the appendix changes both learner and optimization recipe. The paired full-schedule ViT-S results in Table 3 provide the controlled comparison. The effect extends beyond the default pairing, while its magnitude and the DINO result show that it depends on the learner and recipe.

Generation mechanism. Under matched sampling settings, FLUX.1-schnell and SD3 obtain four-task averages of 19.68 and 19.70, so generator identity has little effect once the sampling recipe is controlled. The real-restoration condition returns images removed when constructing the long tail and serves as an oracle rather than a duplication baseline. Its strong result shows that targeted addition remains useful when missing real support is available. Separate duplicate and conventional-augmentation controls test whether reweighting anchors is sufficient.

Sample selection. Table 4 compares three upper-tail mode-window variants with top-tail and uniform sampling. All three diversity-aware mode-window variants are much stronger. The q50–99 and q1–99 variants remain close to the default, showing that the exact lower cutoff is not critical once selection is restricted to a sparse band and diversified within it. We use q75–99 as a fixed default rather than claiming it is universally optimal.

Cycle count. Performance improves sharply over several repair rounds. Relative to 2 cycles, 5 cycles improve the seven-task averages by 7.3 points for linear probing and 5.7 points for kNN, while remaining stronger than 10 cycles. Longer schedules therefore do not help monotonically in this regime.

Table 4: Selection and cycle ablations on ImageNet-100-LT. This matched ablation family is independent of the source-regime runs in Table 1. Values average all seven downstream tasks over three seeds. The nearby mode-window results show that the exact lower percentile is not critical. Uniform and top-tail selection test the dense-allocation and extreme-sparsity alternatives.

Group	Setting	Linear probe	kNN
Selection	Mode-window q1–99	42.1 ± 0.7	33.4 ± 0.3
	Mode-window q50–99	42.2 ± 0.9	33.6 ± 0.3
	Mode-window q75–99	42.3 ± 0.7	33.6 ± 0.5
	Uniform	35.5 ± 1.2	28.4 ± 0.6
	Top-tail	34.3 ± 1.2	27.3 ± 0.6
Schedule	2 cycles	35.0 ± 1.4	27.9 ± 0.5
	5 cycles	42.3 ± 0.7	33.6 ± 0.5
	10 cycles	34.1 ± 1.7	28.1 ± 0.6

Encoder architecture. ResNet-50 gives the strongest seven-task linear-probe average at 42.3 and leads on six targets. ViT-B is best on Flowers but averages 34.8, making ResNet-50 the preferred backbone in this sweep.

kNN corroboration. The appendix reports nonparametric transfer for every ablation. Mode-window selection remains stronger than top-tail and uniform selection over five cycles, and five cycles remains stronger than shorter schedules. Paired ViT and matched FLUX experiments provide the controlled objective and generator comparisons.

6.4 Representation analysis

We use a fixed panel of original images to keep the evaluation population unchanged across checkpoints. With MoCo v3 and ViT-S, effective rank increases by 21.97 ± 1.58, spectral entropy by 0.296 ± 0.019, and 1-NN accuracy by 3.80 ± 0.80. The rank and entropy changes remain significant after Holm correction. SimCLR improves transfer while its global effective rank decreases, so increasing global rank is not a universal account of the method.

We separately test the local intervention in the frozen cycle-0 representation. First-cycle anchors are matched one-to-one within class to untreated images with the same initial kNN radius. After inserting generated samples, the radius falls more around treated anchors in every seed, with a mean difference-in-differences of -0.00134 ± 0.00067. Neighborhood purity changes by $+0.0134 \pm 0.0032$ relative to the matched controls. This verifies that the repair operator places support near the selected regions. It does not by itself establish that subsequent SSL must preserve that local geometry, and the learned-space anchor comparison is mixed for SimCLR. We therefore use the transfer results as the primary measure of utility and report the full fixed-space and learned-space analyses in the appendix.

7 Limitations

BRIDGE relies on two capabilities. The encoder must provide neighborhoods that are informative enough to guide acquisition, and the generator must produce faithful local variants. Failures in either component can direct later cycles toward artifacts. The frozen generator also imports biases from its pretraining data. The score band, neighborhood size, and repair budget are operating choices whose robustness we study empirically rather than universal constants.

Each cycle adds a full embedding pass and a fixed amount of generation. This increases measured wall-clock time by 29–46 percent in our settings. A fixed generation budget has lower relative cost on a larger source, though its effectiveness may eventually require the budget to grow. Our web-source experiments use controlled 10k subsets, so corpus-scale behavior remains an open empirical question.

8 Conclusion

We formulate skewed-data SSL as a feedback problem in which the learner allocates a finite budget to change its future source distribution. Recoverability-weighted support utility links sparse-region repair, interior acquisition, diversity, and adaptive reallocation. BRIDGE realizes these principles through a label-free loop that improves transfer more reliably than uniform addition and maximal-outlier generation across the regimes studied here. Adaptive source repair offers a practical complement to improving the SSL objective alone.

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A Comparison with Related Acquisition Methods

Table 5 makes the comparison axes explicit. Classical oversampling and TADA receive labels or supervised learning signals. DOPING uses latent rarity but targets anomaly detection and does not close the loop through a changing SSL representation. GenRep and StableRep make synthetic images or latent neighborhoods the source of contrastive views without allocating a repair budget over an observed skewed corpus. DiffAug is the closest iterative SSL antecedent, but it jointly learns a generator for generic positive views rather than asking where a frozen generator should intervene. AIDE and industrial data-engine systems establish the broader model–data feedback pattern, while using supervised task failures, VLM/LLM reasoning, annotation, and domain-specific verification. BRIDGE studies the label-free special case in which local support in the current SSL representation is both the diagnostic and acquisition signal. Our VLM rare-concept baseline adapts the AIDE idea by clustering frozen CLIP features, selecting inversely to cluster occupancy, captioning the selected images, and generating from text. The paired text-to-image and captioned SDEdit controls separately test whether retaining the selected image as a local condition matters.

The generator literature also identifies the origins of the repair operator and its controls. SDEdit supplies the image-editing mechanism, while synthetic-recognition studies establish the importance

Table 5: Closest methods and the dimensions relevant to the acquisition problem. “Feedback” means that the learner or task model changes the next acquisition decision.

Method	Labels or task loss	Acquisition signal	Data action	Feedback	Primary goal
SMOTE (Chawla et al., 2002)	class labels	class minority	interpolate examples	no	supervised imbalance
DOPING (Lim et al., 2018)	no labels	latent edge density	latent oversampling	no	anomaly detection
Concept pruning (Abbas et al., 2024)	no task labels	cluster complexity	remove examples	no	efficient pretraining
DA-Fusion (Trabucco et al., 2024)	labels / few-shot set	class and exemplar	image-conditioned diffusion	no	supervised recognition
TADA (Nguyen et al., 2026)	supervised loss dynamics	slow-learned examples	faithful diffusion variants	no	supervised recognition
GenRep / StableRep (Jahanian et al., 2022; Tian et al., 2023)	prompts or generator latent	global sampling	synthetic SSL views	no	representation learning
DiffAug (Zang et al., 2024)	no labels	learned positive-view objective	jointly learned diffusion views	yes	contrastive learning
AIDE (Liang et al., 2024)	task labels and failures	VLM/LLM failure analysis	curate, generate, annotate	yes	driving detection
BRIDGE	no labels	local support in current SSL space	budgeted local variants	yes	source-support repair

Table 6: Local support added in the frozen cycle-0 representation, reported as mean $\pm$ standard deviation over three seeds. Controls are matched one-to-one within class on initial kNN radius and are never selected. Negative radius DiD and positive purity DiD favor treated anchors.

	Δr anchor	Δr control	Radius DiD	Purity DiD
Mean $\pm$ SD	-0.00273 ± 0.00062	-0.00139 ± 0.00004	-0.00134 ± 0.00067	$+0.0134 \pm 0.0032$

of prompts, filtering, diversity, and domain gap. DreamTeacher and DIFT motivate direct generator-feature transfer, and REPA studies the reverse direction from a visual encoder into a diffusion model. Omniverse Replicator provides non-archival systems context for iterative synthetic-data pipelines. BRIDGE brings these tools into a label-free local acquisition problem with a finite intervention budget and repeated re-estimation.

B Representation Diagnostics

The fixed-space intervention test freezes the cycle-0 encoder and queries the same original images before and after inserting all generated samples. First-cycle anchors are matched one-to-one within class to original images with the closest initial kNN radius that are never selected in any cycle. Table 6 reports the direct anchor-versus-control comparison. The learned-space comparison retrains the encoder after the intervention and evaluates the same original-image panel. Its local effect is mixed for SimCLR, while the global response depends on the objective. We therefore treat the fixed-space result as evidence about where new support is inserted, not as evidence that all learned representations must change in the same geometric direction.

C Proofs for the Support-Allocation Framework

Proof of Proposition 1. Sort the distances from z to points in Z and denote their kth order statistic by $r_k(z;Z)$. Adding A preserves every old candidate distance and may introduce smaller ones. The kth order statistic can therefore only decrease. It decreases strictly when an added distance enters the first k positions and displaces a strictly larger distance.

Proof of Proposition 2. Let $x = n_j + m_j + \kappa$. Since $f(x) = x^{-\gamma}$ is positive, strictly decreasing, and strictly convex for $x > 0$, the forward difference $f(x) - f(x+1)$ is positive and strictly decreases with x. Multiplication by $p_j a_j > 0$ preserves both properties. Equal $p_j a_j$ therefore leaves effective support as the sole determinant of the marginal ordering.

Proof of Proposition 3. Rewrite the utility as $u(d) = \rho(d)(b(d) + \lambda) - \lambda$. Its derivative is

$$u'(d) = \rho(d)(b(d) + \lambda)\left[\frac{b'(d)}{b(d) + \lambda} + \frac{\rho'(d)}{\rho(d)}\right] = \rho(d)(b(d) + \lambda)g(d).$$

The factor preceding $g(d)$ is positive. Since g is strictly decreasing and changes sign across the interval, it has one root $d^\star$ by continuity. Hence u increases before $d^\star$ and decreases after it, making $d^\star$ the unique global maximizer.

Proof of Proposition 4. For region j, define the marginal sequence $\delta_{j,t} = U_j(n_j+t) - U_j(n_j+t-1)$ for $t \geq 1$. Discrete concavity makes each sequence nonincreasing. Any feasible allocation of B

repairs selects a prefix of length m_j from each sequence. Greedy allocation repeatedly chooses the largest available next element. After B steps it has selected the B largest prefix-feasible marginals. If another allocation had larger utility, it would contain a selected marginal smaller than an available unselected marginal, and exchanging them would improve that allocation. This contradicts optimality. A frozen allocation can fail this rule because it preserves the initial ordering after selected regions move to smaller marginals.

D Implementation and Reproducibility

Core training configuration. The main experiments use ResNet-50 encoders. Most baseline, BRIDGE, and source-regime comparisons use 5 cycles with 100 epochs per cycle. The default BRIDGE configuration uses mean kNN distance with $k = 100$, selects 500 samples per cycle, and generates 5 images per selected sample. The default optimizer is AdamW with weight decay 10^{-4}, and method-specific learning rates and temperatures follow the settings used in each experiment.

OOD scoring and selection. The submitted experiments used raw squared ℓ_2 distance for OOD scoring and normalized features for farthest-point sampling. Because raw feature norms can confound cross-objective comparisons, the new controlled experiments use normalized ℓ_2 throughout and report raw ℓ_2, normalized ℓ_2, cosine, and multiscale-k variants as a robustness study. The mode-window selector restricts the OOD distribution to the 75th to 99th percentile range, builds a histogram with the code-level auto-bin heuristic, forms a mode-centered candidate pool of size $4B$, and then applies greedy farthest-point sampling to select the final B anchors.

Generators. The main method uses Stable Diffusion 3 in image-to-image mode. The generation ablation replaces it with FLUX or with a no-generation re-population baseline. In the PASS and DiffusionDB experiments, we increase the SD3 batch size to 12 to fit the 24-hour wall-time limit.

Source data specifics. PASS and DiffusionDB use `train[:10000]` with pretraining split override `[1.0, 0.0, 0.0]`. This keeps the source data budget aligned across methods while avoiding unnecessary pretraining validation overhead. The long-tailed ImageNet-100, CIFAR-10, and CIFAR-100 source regimes use their standard long-tailed source configurations, without the 10k subset restriction used for PASS and DiffusionDB.

Seeds and reporting. All main claims are supported by three seed averages. Tables report mean $\pm$ standard deviation over those seeds. The exact table values are computed directly from the three runs corresponding to each reported setting.

Code availability. An anonymous code release accompanying this submission is available at `https://anonymous.4open.science/status/FOMO-0488`.

E Compute Resources and Runtime

Most reported experiments use one node with two NVIDIA A100 80GB GPUs and a 24-hour wall-time limit. The DINO ablation uses four A100 80GB GPUs after reducing the number of local crops from six to two. Across the successful experiments we report, the aggregate training time is approximately 3022 wall-clock hours, or about 6142 GPU hours. This excludes failed runs, timeout runs, cache migration jobs, and other debugging or pilot jobs, so the full project required more compute than the successful totals alone.

Representative mean wall-clock times over three seeds are as follows. On ImageNet-100-LT source pretraining, SimCLR takes 16.9 hours and BRIDGE takes 21.8 hours. On PASS, SimCLR takes 10.2 hours and BRIDGE takes 14.9 hours. On DiffusionDB, SimCLR takes 11.6 hours and BRIDGE takes 16.4 hours. The successful DINO ablation takes 16.3 hours on four GPUs after reducing the number of local crops from six to two.

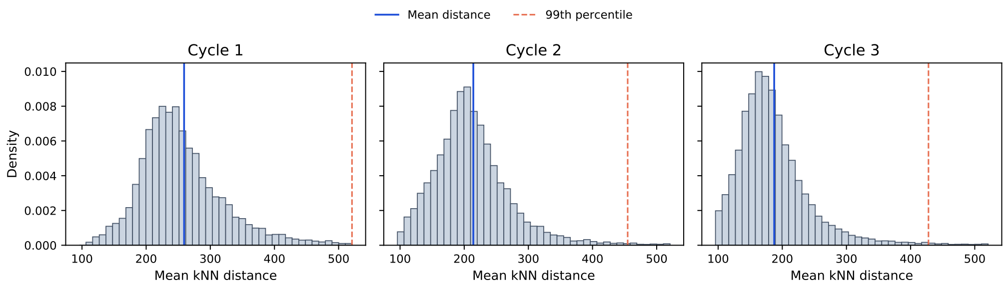

Figure 2: Representative OOD histograms across BRIDGE cycles from an ImageNet-100-LT run. Solid blue lines mark the mean distance and dashed orange lines mark the 99th percentile. The acquisition distribution and its modal shoulder change as training and repair progress.

F ASSETS AND TERMS OF USE

We use existing datasets and pretrained generators under their original release terms.

Source datasets. ImageNet-100-LT is derived from ImageNet, whose official access terms restrict the images to non commercial research and educational use. PASS is released under Creative Commons Attribution 4.0. DiffusionDB is released under CC0-1.0. CIFAR-10, CIFAR-100, Stanford Cars, FGVC Aircraft, Oxford Flowers, and Oxford-IIIT Pets are used from their official academic benchmark releases. When an official page provides download terms instead of a standard software style license, we follow those original terms.

Generators. Stable Diffusion 3 Medium is used under the Stability AI Community License. FLUX.1-dev is used only in the ablation and is subject to the FLUX.1-dev Non-Commercial License. We do not redistribute model weights, third party datasets, or generated image corpora as part of this submission.

G BROADER IMPACT

A method that improves representation quality under biased source data can reduce the need for manual relabeling and can make SSL more useful in domains where balanced data are difficult to collect. That is a positive outcome. At the same time, our method relies on large image generators and on web-sourced datasets. These carry familiar risks related to license compliance, privacy, and unintended memorization or harmful content. Our experiments stay within existing dataset and model release terms and do not release new generators or scraped image collections. Any future deployment would still require a task-specific review of fairness, privacy, and content safety.

H Full Results

Table 7: Full linear-probe transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 ± 0.5	46.8 ± 0.4	13.7 ± 0.3	19.6 ± 0.2	35.9 ± 7.2	38.6 ± 1.5	50.4 ± 0.5	39.8 ± 1.5
	TS	72.5 ± 0.4	44.0 ± 0.5	12.5 ± 1.2	15.0 ± 0.1	35.7 ± 1.8	40.6 ± 2.2	**56.7 ± 1.2**	39.6 ± 1.1
	SDCLR	41.1 ± 1.7	18.0 ± 0.6	1.4 ± 0.3	2.3 ± 0.5	10.8 ± 1.7	10.4 ± 0.3	21.9 ± 0.6	15.1 ± 0.8
	BRIDGE (ours)	**75.8 ± 0.2**	**49.3 ± 0.2**	**16.4 ± 1.2**	**22.4 ± 0.4**	**36.8 ± 1.9**	**42.9 ± 1.0**	53.8 ± 0.1	**42.5 ± 0.7**
	BRIDGE+TS (ours)	74.2 ± 0.5	45.6 ± 0.8	9.4 ± 0.5	15.0 ± 0.2	25.5 ± 2.5	33.1 ± 2.8	48.3 ± 0.4	35.9 ± 1.1
	BRIDGE+SDCLR (ours)	43.8 ± 1.7	18.4 ± 0.7	2.0 ± 0.3	2.5 ± 0.8	10.0 ± 0.8	12.2 ± 3.1	21.8 ± 0.5	15.8 ± 1.1
CIFAR-10-LT	SimCLR	67.8 ± 0.2	38.0 ± 1.0	6.5 ± 0.4	16.0 ± 0.9	**19.2 ± 1.6**	24.0 ± 1.3	32.0 ± 0.7	29.1 ± 0.9
	TS	68.5 ± 1.3	36.0 ± 1.7	4.3 ± 0.2	10.3 ± 0.1	13.1 ± 2.0	19.7 ± 2.3	31.5 ± 1.2	26.2 ± 1.3
	SDCLR	39.6 ± 1.1	15.6 ± 0.2	1.1 ± 0.1	2.4 ± 0.9	12.0 ± 1.6	9.1 ± 1.8	16.5 ± 1.6	13.8 ± 1.1
	BRIDGE (ours)	78.1 ± 0.3	**49.8 ± 0.4**	**9.2 ± 0.8**	**17.7 ± 1.0**	18.2 ± 1.1	**27.4 ± 2.7**	39.0 ± 0.6	**34.2 ± 1.0**
	BRIDGE+TS (ours)	**78.6 ± 0.3**	46.1 ± 0.3	7.5 ± 0.9	11.1 ± 0.7	14.5 ± 0.6	25.1 ± 2.7	**39.6 ± 0.8**	31.8 ± 0.9
	BRIDGE+SDCLR (ours)	40.9 ± 0.7	17.9 ± 0.5	1.4 ± 0.1	2.2 ± 0.6	11.0 ± 2.4	10.0 ± 2.7	21.0 ± 0.6	14.9 ± 1.1
CIFAR-100-LT	SimCLR	59.0 ± 1.5	31.1 ± 1.6	4.1 ± 0.3	11.2 ± 1.0	12.6 ± 3.4	20.1 ± 1.0	28.7 ± 1.3	23.8 ± 1.4
	TS	56.3 ± 0.2	28.0 ± 0.1	2.8 ± 0.6	8.5 ± 0.2	15.5 ± 2.5	16.2 ± 3.1	26.2 ± 1.8	21.9 ± 1.2
	SDCLR	37.1 ± 0.8	15.7 ± 0.2	1.9 ± 0.1	2.4 ± 0.7	9.4 ± 1.0	9.9 ± 1.7	16.2 ± 0.3	13.2 ± 0.7
	BRIDGE (ours)	**74.2 ± 0.5**	**48.1 ± 0.4**	**8.3 ± 1.0**	**18.2 ± 0.7**	**24.4 ± 2.3**	27.6 ± 0.7	39.8 ± 1.1	**34.4 ± 1.0**
	BRIDGE+TS (ours)	73.7 ± 0.1	47.3 ± 0.5	7.9 ± 0.6	14.1 ± 0.5	18.2 ± 3.2	**28.4 ± 0.4**	**41.3 ± 0.3**	33.0 ± 0.8
	BRIDGE+SDCLR (ours)	42.0 ± 1.5	16.9 ± 0.9	1.3 ± 0.1	2.2 ± 0.6	10.6 ± 0.8	8.3 ± 0.8	20.2 ± 0.9	14.5 ± 0.8
PASS-10k	SimCLR	72.3 ± 0.2	45.5 ± 0.5	11.0 ± 0.2	17.5 ± 1.4	33.3 ± 1.3	33.3 ± 0.9	44.7 ± 0.9	36.8 ± 0.8
	TS	71.5 ± 0.2	43.5 ± 0.4	10.2 ± 1.2	17.2 ± 0.2	33.1 ± 2.1	36.4 ± 1.8	49.4 ± 0.4	37.3 ± 0.9
	SDCLR	42.2 ± 1.1	17.7 ± 0.8	1.8 ± 0.3	2.2 ± 0.8	10.6 ± 1.3	8.2 ± 1.6	20.8 ± 0.4	14.8 ± 0.9
	BRIDGE (ours)	**73.9 ± 0.3**	**47.1 ± 0.4**	**13.6 ± 0.7**	**20.5 ± 0.7**	**37.7 ± 4.0**	**37.1 ± 0.7**	49.1 ± 0.5	**39.9 ± 1.0**
	BRIDGE+TS (ours)	68.9 ± 0.6	40.6 ± 0.7	12.1 ± 0.4	14.1 ± 0.5	34.6 ± 2.9	31.6 ± 1.5	**52.4 ± 0.6**	36.4 ± 1.0
	BRIDGE+SDCLR (ours)	44.0 ± 0.9	18.8 ± 0.9	1.4 ± 0.3	1.8 ± 0.9	12.6 ± 2.6	12.4 ± 1.3	21.2 ± 1.1	16.0 ± 1.1
DiffusionDB-10k	SimCLR	74.4 ± 0.5	48.6 ± 0.2	10.9 ± 0.7	17.6 ± 0.8	31.2 ± 2.4	36.1 ± 1.3	45.3 ± 0.5	37.7 ± 0.9
	TS	72.3 ± 0.3	44.4 ± 0.3	10.7 ± 0.6	15.4 ± 0.6	33.1 ± 1.1	**39.9 ± 2.7**	48.8 ± 0.2	37.8 ± 0.8
	SDCLR	42.4 ± 0.2	18.4 ± 0.2	1.1 ± 0.2	2.4 ± 0.1	13.0 ± 3.7	8.2 ± 1.4	21.0 ± 0.6	15.2 ± 0.9
	BRIDGE (ours)	**75.9 ± 0.4**	**49.9 ± 0.5**	**12.8 ± 1.0**	**20.0 ± 0.4**	**35.7 ± 1.1**	37.7 ± 0.8	49.1 ± 0.5	**40.1 ± 0.7**
	BRIDGE+TS (ours)	67.8 ± 0.6	39.1 ± 0.5	9.4 ± 1.2	12.4 ± 0.4	30.3 ± 3.0	33.7 ± 1.9	**50.8 ± 0.6**	34.8 ± 1.2
	BRIDGE+SDCLR (ours)	43.8 ± 0.9	18.5 ± 0.3	1.3 ± 0.4	2.5 ± 0.0	14.3 ± 1.6	11.4 ± 0.6	21.9 ± 0.6	16.2 ± 0.6

Table 8: Full kNN transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 ± 0.4	37.3 ± 0.2	6.0 ± 0.7	11.2 ± 0.4	32.2 ± 0.7	23.6 ± 0.3	43.9 ± 0.9	31.8 ± 0.5
	TS	68.4 ± 0.3	36.5 ± 0.1	4.7 ± 0.2	8.3 ± 0.5	26.6 ± 1.0	23.6 ± 1.7	**52.2 ± 0.5**	31.5 ± 0.6
	SDCLR	32.5 ± 0.4	10.9 ± 0.2	1.0 ± 0.3	2.1 ± 0.1	10.7 ± 0.7	4.7 ± 0.2	11.2 ± 0.2	10.4 ± 0.3
	BRIDGE (ours)	**69.0 ± 0.5**	**37.8 ± 0.3**	**7.0 ± 0.2**	**11.7 ± 0.2**	**36.5 ± 0.4**	**25.3 ± 1.0**	48.1 ± 0.6	**33.6 ± 0.4**
	BRIDGE+TS (ours)	65.4 ± 0.2	33.6 ± 0.3	3.1 ± 0.1	7.1 ± 0.3	18.2 ± 0.2	15.6 ± 0.7	40.8 ± 0.5	26.3 ± 0.3
	BRIDGE+SDCLR (ours)	36.3 ± 0.8	13.0 ± 0.5	1.1 ± 0.3	2.4 ± 0.2	9.7 ± 0.3	5.4 ± 0.4	13.9 ± 1.0	11.7 ± 0.5
CIFAR-10-LT	SimCLR	66.9 ± 0.4	33.4 ± 0.3	3.7 ± 0.4	9.8 ± 0.7	16.5 ± 0.8	13.9 ± 0.8	24.0 ± 0.8	24.0 ± 0.6
	TS	66.3 ± 0.6	27.8 ± 0.4	2.4 ± 0.6	6.4 ± 0.2	10.5 ± 0.7	11.1 ± 0.4	22.0 ± 0.9	20.9 ± 0.5
	SDCLR	30.1 ± 0.3	10.0 ± 0.3	1.7 ± 0.3	2.2 ± 0.2	10.1 ± 0.7	4.4 ± 0.4	9.8 ± 0.3	9.7 ± 0.4
	BRIDGE (ours)	74.8 ± 0.4	**41.0 ± 0.4**	**4.8 ± 0.5**	**10.0 ± 0.2**	**24.2 ± 0.6**	**16.2 ± 0.7**	**29.7 ± 0.3**	**28.7 ± 0.4**
	BRIDGE+TS (ours)	**77.6 ± 0.7**	35.3 ± 0.5	3.1 ± 0.3	6.6 ± 0.4	16.3 ± 0.6	12.5 ± 1.0	26.7 ± 0.1	25.5 ± 0.5
	BRIDGE+SDCLR (ours)	34.4 ± 1.2	11.4 ± 0.6	1.2 ± 0.2	2.2 ± 0.2	8.8 ± 1.5	5.0 ± 0.2	11.7 ± 1.1	10.7 ± 0.7
CIFAR-100-LT	SimCLR	59.3 ± 0.8	30.1 ± 0.9	3.0 ± 0.3	7.4 ± 0.4	13.7 ± 0.2	10.8 ± 0.4	21.8 ± 1.1	20.9 ± 0.6
	TS	53.1 ± 0.6	24.6 ± 0.4	2.4 ± 0.4	5.5 ± 0.3	9.3 ± 0.4	8.3 ± 0.2	18.7 ± 1.0	17.4 ± 0.5
	SDCLR	25.7 ± 1.2	8.5 ± 0.6	0.9 ± 0.4	1.9 ± 0.2	7.3 ± 0.3	3.9 ± 0.3	8.3 ± 1.1	8.1 ± 0.6
	BRIDGE (ours)	**69.3 ± 0.5**	**39.5 ± 0.1**	**4.8 ± 0.6**	**8.9 ± 0.1**	**21.9 ± 0.1**	**15.5 ± 0.1**	**29.1 ± 0.7**	**27.0 ± 0.3**
	BRIDGE+TS (ours)	64.3 ± 0.4	36.6 ± 0.4	2.9 ± 0.3	6.3 ± 0.1	16.4 ± 0.5	12.3 ± 0.0	28.2 ± 0.6	23.9 ± 0.3
	BRIDGE+SDCLR (ours)	31.9 ± 0.4	10.4 ± 0.5	0.8 ± 0.1	2.1 ± 0.5	7.5 ± 1.6	4.5 ± 0.1	9.1 ± 0.7	9.5 ± 0.6
PASS-10k	SimCLR	67.2 ± 0.2	**37.0 ± 0.3**	5.4 ± 0.3	10.3 ± 0.5	30.4 ± 0.0	16.7 ± 0.5	38.3 ± 0.3	29.3 ± 0.3
	TS	65.2 ± 0.7	34.6 ± 0.3	4.4 ± 0.2	7.9 ± 0.2	24.5 ± 0.1	14.4 ± 0.4	39.3 ± 0.4	27.2 ± 0.3
	SDCLR	32.4 ± 1.5	11.5 ± 0.8	1.3 ± 0.2	2.0 ± 0.1	8.6 ± 0.7	5.2 ± 0.2	11.1 ± 0.8	10.3 ± 0.6
	BRIDGE (ours)	**67.5 ± 0.3**	36.9 ± 0.3	**5.8 ± 0.5**	**10.7 ± 0.4**	**33.7 ± 0.2**	**17.8 ± 0.6**	**41.4 ± 0.8**	**30.6 ± 0.4**
	BRIDGE+TS (ours)	58.8 ± 0.5	27.9 ± 0.5	3.0 ± 0.2	6.6 ± 0.1	23.0 ± 0.3	13.4 ± 0.4	38.8 ± 0.8	24.5 ± 0.4
	BRIDGE+SDCLR (ours)	37.1 ± 2.0	13.6 ± 1.0	1.6 ± 0.2	2.4 ± 0.2	10.1 ± 0.5	6.3 ± 0.3	13.1 ± 0.6	12.0 ± 0.7
DiffusionDB-10k	SimCLR	67.3 ± 0.1	36.2 ± 0.4	4.3 ± 0.2	8.8 ± 0.1	26.8 ± 0.4	17.3 ± 0.1	36.7 ± 0.8	28.2 ± 0.3
	TS	64.7 ± 0.4	33.6 ± 0.3	3.1 ± 0.5	7.3 ± 0.3	23.2 ± 0.4	17.0 ± 0.0	37.8 ± 0.6	26.7 ± 0.4
	SDCLR	32.6 ± 1.2	10.0 ± 0.3	1.3 ± 0.1	2.2 ± 0.2	6.1 ± 0.7	4.8 ± 0.4	10.6 ± 0.5	9.7 ± 0.5
	BRIDGE (ours)	**68.3 ± 0.2**	**37.4 ± 0.4**	**5.4 ± 0.3**	**9.8 ± 0.1**	**30.4 ± 0.2**	**18.6 ± 0.2**	**39.4 ± 1.0**	**29.9 ± 0.3**
	BRIDGE+TS (ours)	57.5 ± 0.8	26.6 ± 0.3	3.0 ± 0.4	6.5 ± 0.1	21.7 ± 0.8	15.5 ± 0.4	37.0 ± 0.4	24.0 ± 0.5
	BRIDGE+SDCLR (ours)	35.2 ± 0.7	11.1 ± 0.3	1.0 ± 0.1	2.2 ± 0.1	8.1 ± 1.0	5.0 ± 0.3	11.3 ± 0.7	10.6 ± 0.5

Table 9: Pretraining objective ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR (default)	**76.0 ± 0.3**	**49.1 ± 0.2**	**15.2 ± 1.2**	**22.4 ± 0.4**	**36.8 ± 1.9**	**42.9 ± 1.0**	**53.8 ± 0.1**	**42.3 ± 0.7**
MoCo	50.5 ± 1.0	20.6 ± 2.7	3.0 ± 0.2	5.7 ± 0.5	11.5 ± 3.0	15.3 ± 0.3	28.2 ± 0.7	19.3 ± 1.2
DINO	43.5 ± 1.6	18.3 ± 1.0	1.8 ± 0.7	1.7 ± 0.1	12.4 ± 3.6	9.1 ± 1.1	20.3 ± 0.6	15.3 ± 1.2

Table 10: Pretraining objective ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR (default)	**69.0 ± 0.5**	**37.8 ± 0.3**	**6.7 ± 0.3**	**11.7 ± 0.2**	**36.5 ± 0.4**	**25.3 ± 1.0**	**48.1 ± 0.6**	**33.6 ± 0.5**
MoCo	58.0 ± 0.4	26.7 ± 0.5	3.0 ± 0.2	5.4 ± 0.2	16.3 ± 0.9	12.2 ± 0.6	25.1 ± 0.5	21.0 ± 0.5
DINO	35.8 ± 0.4	12.7 ± 0.1	1.6 ± 0.4	2.2 ± 0.2	6.0 ± 0.5	5.8 ± 0.2	11.2 ± 0.4	10.7 ± 0.3

Table 11: Augmentation mechanism ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	**75.0 ± 0.3**	**48.5 ± 0.2**	**16.3 ± 0.8**	**20.7 ± 0.4**	35.7 ± 0.9	**41.8 ± 0.8**	**52.4 ± 0.7**	**41.5 ± 0.6**
FLUX	70.7 ± 0.4	42.1 ± 1.4	10.3 ± 1.3	17.5 ± 0.9	29.4 ± 4.6	31.5 ± 2.9	43.7 ± 1.4	35.0 ± 1.9
No-generation re-population	74.8 ± 0.4	47.8 ± 0.7	13.1 ± 0.1	20.5 ± 0.5	**40.3 ± 6.1**	38.1 ± 1.5	50.3 ± 0.4	40.7 ± 1.4

Table 12: Augmentation mechanism ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	68.6 ± 0.1	37.2 ± 0.2	6.4 ± 0.6	**11.5 ± 0.2**	**35.2 ± 0.3**	**24.3 ± 0.6**	**46.8 ± 0.7**	**32.9 ± 0.4**
FLUX	65.8 ± 1.0	35.4 ± 0.8	5.7 ± 0.7	9.4 ± 0.9	27.7 ± 1.0	18.8 ± 1.3	38.7 ± 1.2	28.8 ± 1.0
No-generation re-population	**68.8 ± 0.2**	**37.7 ± 0.3**	**6.6 ± 0.5**	11.4 ± 0.5	32.7 ± 0.1	23.6 ± 0.6	44.4 ± 0.4	32.2 ± 0.4

Table 13: Selection rule ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	75.6 ± 0.1	49.2 ± 0.2	**16.9 ± 0.7**	21.9 ± 0.4	34.6 ± 2.9	43.5 ± 0.7	52.8 ± 0.1	42.1 ± 0.7
Mode-window q50-99 + diversity	75.5 ± 0.1	**49.3 ± 0.4**	15.2 ± 0.3	**22.4 ± 0.6**	35.5 ± 1.1	**43.8 ± 3.1**	53.4 ± 0.9	42.2 ± 0.9
Mode-window q75-99 + diversity (default)	**76.0 ± 0.3**	49.1 ± 0.2	15.2 ± 1.2	22.4 ± 0.4	**36.8 ± 1.9**	42.9 ± 1.0	**53.8 ± 0.1**	**42.3 ± 0.7**
Top-tail	70.8 ± 0.5	43.1 ± 1.0	10.0 ± 1.7	15.9 ± 1.3	27.7 ± 1.0	30.5 ± 1.2	42.2 ± 1.4	34.3 ± 1.2
Uniform	71.5 ± 0.2	43.8 ± 0.7	9.3 ± 0.9	17.5 ± 1.2	32.9 ± 0.4	31.7 ± 3.9	42.0 ± 0.9	35.5 ± 1.2

Table 14: Selection rule ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	**69.2 ± 0.3**	37.7 ± 0.2	6.6 ± 0.2	11.4 ± 0.2	35.9 ± 0.2	25.0 ± 0.5	48.3 ± 0.3	33.4 ± 0.3
Mode-window q50-99 + diversity	68.7 ± 0.1	**38.0 ± 0.2**	**7.3 ± 0.2**	11.3 ± 0.2	**36.5 ± 0.1**	24.8 ± 0.8	**48.5 ± 0.4**	33.6 ± 0.3
Mode-window q75-99 + diversity (default)	69.0 ± 0.5	37.8 ± 0.3	6.7 ± 0.3	**11.7 ± 0.2**	36.5 ± 0.4	**25.3 ± 1.0**	48.1 ± 0.6	**33.6 ± 0.5**
Top-tail	65.7 ± 0.8	34.4 ± 0.2	4.1 ± 0.4	9.6 ± 0.2	23.9 ± 0.9	18.1 ± 0.8	35.4 ± 1.2	27.3 ± 0.6
Uniform	66.1 ± 1.0	35.7 ± 0.3	4.9 ± 0.1	10.2 ± 0.7	25.3 ± 0.4	19.0 ± 0.8	37.4 ± 0.6	28.4 ± 0.6

Table 15: Cycle count ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	68.9 ± 0.6	41.6 ± 1.3	10.6 ± 1.2	16.4 ± 0.4	30.7 ± 4.5	33.2 ± 0.9	43.3 ± 0.6	35.0 ± 1.4
5 cycles (default)	**76.0 ± 0.3**	**49.1 ± 0.2**	**15.2 ± 1.2**	**22.4 ± 0.4**	**36.8 ± 1.9**	**42.9 ± 1.0**	**53.8 ± 0.1**	**42.3 ± 0.7**
10 cycles	70.5 ± 1.2	43.8 ± 0.6	9.2 ± 0.9	17.2 ± 1.6	24.8 ± 5.2	31.3 ± 1.7	41.9 ± 0.6	34.1 ± 1.7

Table 16: Cycle count ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	65.3 ± 0.4	34.2 ± 0.5	5.0 ± 0.7	10.0 ± 0.2	26.1 ± 0.8	18.7 ± 0.6	35.9 ± 0.3	27.9 ± 0.5
5 cycles (default)	**69.0 ± 0.5**	**37.8 ± 0.3**	**6.7 ± 0.3**	**11.7 ± 0.2**	**36.5 ± 0.4**	**25.3 ± 1.0**	**48.1 ± 0.6**	**33.6 ± 0.5**
10 cycles	66.4 ± 0.5	35.4 ± 0.3	4.8 ± 0.3	10.4 ± 0.6	25.2 ± 0.8	18.6 ± 0.9	36.0 ± 0.7	28.1 ± 0.6

Table 17: Backbone architecture ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	68.8 ± 0.1	41.7 ± 0.2	13.5 ± 0.3	19.7 ± 0.6	38.7 ± 3.4	38.4 ± 2.6	50.2 ± 0.2	38.7 ± 1.1
ResNet-50 (default)	**76.0 ± 0.3**	**49.1 ± 0.2**	**15.2 ± 1.2**	**22.4 ± 0.4**	36.8 ± 1.9	**42.9 ± 1.0**	**53.8 ± 0.1**	**42.3 ± 0.7**
ViT-S	62.7 ± 0.4	37.6 ± 0.6	8.8 ± 0.5	12.2 ± 0.2	35.1 ± 4.2	30.2 ± 1.6	42.5 ± 1.2	32.7 ± 1.3
ViT-B	66.0 ± 2.3	40.6 ± 2.3	8.3 ± 0.5	14.3 ± 0.8	**38.8 ± 2.6**	31.3 ± 2.1	44.5 ± 1.7	34.8 ± 1.8

Table 18: Backbone architecture ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	66.3 ± 0.2	35.7 ± 0.3	5.9 ± 0.2	11.1 ± 0.4	33.5 ± 0.1	23.3 ± 0.5	44.5 ± 0.7	31.5 ± 0.4
ResNet-50 (default)	69.0 ± 0.5	37.8 ± 0.3	**6.7 ± 0.3**	**11.7 ± 0.2**	36.5 ± 0.4	**25.3 ± 1.0**	**48.1 ± 0.6**	**33.6 ± 0.5**
ViT-S	67.3 ± 0.4	37.5 ± 0.3	5.3 ± 0.2	7.6 ± 0.1	36.2 ± 0.7	20.1 ± 1.0	42.1 ± 1.0	30.9 ± 0.5
ViT-B	**69.3 ± 0.8**	**40.1 ± 0.8**	6.0 ± 0.2	8.3 ± 0.2	**39.8 ± 0.8**	21.9 ± 1.4	45.1 ± 0.7	32.9 ± 0.7

I ALGORITHM

Algorithm 1 BRIDGE training loop with mode-window selection

Initialize source set $D^{(0)}$, encoder $f_{\theta^{(0)}}$, frozen generator g_ϕ, cycle budgets $\{E_c\}_{c=0}^{C-1}$
for $c = 0, \ldots, C-1$ **do**
  Train or continue training $f_{\theta^{(c)}}$ on $D^{(c)}$ for E_c epochs
  Compute embeddings $Z^{(c)} = \{f_{\theta^{(c)}}(x_i) : x_i \in D^{(c)}\}$
  Compute mean kNN scores $\{d_i^{(c)}\}_{i=1}^{N^{(c)}}$ from $Z^{(c)}$
  Restrict $\{d_i^{(c)}\}$ to the 75th to 99th percentile range and build a histogram
  Let $m^{(c)}$ be the center of the modal histogram bin inside that upper-tail band
  Form a candidate set $C^{(c)}$ of size $4B$ whose scores are closest to $m^{(c)}$
  Select $S^{(c)} \subseteq C^{(c)}$ with $|S^{(c)}| = B$ by greedy farthest-point sampling in embedding space
  if $c < C-1$ **then**
    Form $\widetilde{D}^{(c)} = \{g_\phi(x_i, \epsilon_j) \mid i \in S^{(c)},\ j \in [G]\}$
    Update $D^{(c+1)} = D^{(c)} \cup \widetilde{D}^{(c)}$
  end if
end for
Return the final encoder and repaired source set